

Technology Career Questions and Answers

Frequently asked career questions with direct answers, covering how to choose a path, what to learn, how to prepare for interviews and how to change careers.

Choosing a Career Path

Question: What technology career should I choose if I am a complete beginner?

Answer: If you are a complete beginner, the easiest entry points into technology are IT Support Engineer, Data Analyst and QA Engineer, because they need less coding at the start. If you enjoy coding, start with Full Stack Developer. If you enjoy hardware and infrastructure, start with Network Engineer. If you enjoy maths and experiments, aim for Data Analyst first and then Data Scientist or AI Engineer.

Question: What is the difference between an AI Engineer and a Data Scientist?

Answer: An AI Engineer builds and ships AI-powered software products, focusing on models, APIs, deployment and reliability. A Data Scientist analyses data to find insights and answer business questions, focusing on statistics, experiments and communication. The AI Engineer writes more production code, while the Data Scientist writes more analysis and reports.

Question: What is the difference between an AI Engineer and a Machine Learning Engineer?

Answer: The two roles overlap a lot. A Machine Learning Engineer usually trains models from data and manages the training pipeline. An AI Engineer usually takes existing models, especially large language models, and builds applications around them such as chatbots, RAG systems and agents. In smaller companies one person often does both jobs.

Question: What is the difference between a Network Engineer and a Cybersecurity Analyst?

Answer: A Network Engineer builds and maintains the network so data flows correctly. A Cybersecurity Analyst protects systems from attacks and investigates suspicious activity. Networking knowledge is the foundation for both, and many Cybersecurity Analysts start as Network Engineers.

Question: What is the difference between a Network Engineer and a Cloud Engineer?

Answer: A Network Engineer traditionally works with physical routers, switches and firewalls in offices and data centres. A Cloud Engineer works with virtual networks, servers and services inside AWS, Azure or Google Cloud. Many Network Engineers now learn cloud networking and move into cloud roles.

Question: Do I need a computer science degree to work in technology?

Answer: A computer science degree helps, especially for campus hiring and for roles at large companies, but it is not always required. For Full Stack Developer, Data Analyst, IT Support Engineer and QA Engineer roles, a strong project portfolio, certifications and practical skills are often enough. For research roles a degree is usually necessary.

Question: Which technology career pays the most?

Answer: At senior levels, AI Engineer, Machine Learning Engineer, Cloud Architect and specialised Cybersecurity roles are typically the highest paid. However, pay depends more on your experience level, the company and your location than on the job title alone.

Question: Which technology role is easiest to enter without experience?

Answer: IT Support Engineer is usually the easiest technology role to enter with no experience, followed by Data Analyst and QA Engineer. These roles are often used as a first step into networking, cloud, security or engineering.

Skills and Learning

Question: Which programming language should I learn first?

Answer: For AI, data and automation, learn Python first because it is simple and has the largest set of libraries. For web development, learn JavaScript. For large enterprise systems, learn Java. For network automation, Python is the standard choice. Learning one language well matters more than learning several badly.

Question: How long does it take to become job ready?

Answer: With consistent study of about 10 to 15 hours per week, most people become job ready for an entry-level role in about 6 to 12 months. A Data Analyst or IT Support path can be faster, around 4 to 6 months. AI Engineer and Machine Learning Engineer usually take longer because they need programming, maths and deployment skills together.

Question: How important are projects compared to certifications?

Answer: Projects show that you can actually build something and are usually more persuasive in interviews for developer and AI roles. Certifications are more valuable in networking, cloud and cybersecurity, where employers use them to filter candidates. The strongest candidates have both.

Question: What kind of projects should I build for an AI Engineer role?

Answer: Good AI Engineer portfolio projects include a RAG chatbot that answers questions from your own PDF documents, a document summariser, a resume screening tool, a semantic search engine over a dataset, and a small agent that uses tools. Deploy at least one project publicly and write a short README explaining the design choices.

Question: What should I practise for a Network Engineer role?

Answer: For a Network Engineer role, practise subnetting until it becomes automatic, build labs in Cisco Packet Tracer or GNS3, configure VLANs, routing with OSPF, NAT and access control lists, and learn to troubleshoot with ping, traceroute and Wireshark. Studying for the CCNA gives you a good structure.

Question: Do I need to be good at maths for AI?

Answer: For an AI Engineer who builds applications on top of existing models, you need comfortable basic maths: statistics, probability and a general understanding of vectors. For a Machine Learning Engineer or researcher who designs models, stronger linear algebra, calculus and statistics are important.

Question: What are the most in-demand skills right now?

Answer: Widely demanded skills across technology include Python, SQL, cloud platforms such as AWS or Azure, Docker and Kubernetes, and knowledge of large language models and RAG. For infrastructure roles, networking fundamentals and Linux remain essential.

Job Search and Interviews

Question: How should I write a resume for a technical role?

Answer: Keep a technical resume to one page for early career. Put a short summary at the top, then skills grouped by category, then projects with what you built and the result, then experience and education. Use action words and numbers, for example reduced report generation time by 40 percent. Match the keywords in the job description because many companies filter resumes automatically.

Question: What questions are asked in an AI Engineer interview?

Answer: Common AI Engineer interview questions include: explain how RAG works and when you would use it instead of fine-tuning; what is an embedding; how do you reduce hallucination; how do you evaluate the quality of a language model's answers; how would you deploy and monitor a model in production; and Python coding questions on data manipulation.

Question: What questions are asked in a Network Engineer interview?

Answer: Common Network Engineer interview questions include: explain the OSI model layers; what is the difference between TCP and UDP; explain what happens when you type a website address in a browser; how do you subnet a given network; what is the difference between a router and a switch; how would you troubleshoot a user who cannot reach the internet; and explain NAT, DHCP and DNS.

Question: What questions are asked in a Data Analyst interview?

Answer: Common Data Analyst interview questions include SQL joins and group-by queries, how you would handle missing data, explaining a dashboard you built, interpreting an A/B test result, and a case question about why a business metric dropped.

Question: How do I answer 'tell me about yourself' in a technical interview?

Answer: Answer in about 60 to 90 seconds with three parts: who you are now and your background, one or two relevant things you have built or done with a concrete result, and why you are interested in this specific role. Avoid repeating your whole resume line by line.

Question: How do I get experience when every job asks for experience?

Answer: Build and publish real projects, contribute to open-source repositories, take freelance or volunteer work for small organisations, do internships, and take entry roles like IT support or QA that put you inside a technology team. Documented personal projects count as evidence of skill even without a job title.

Question: Should I specialise or stay a generalist?

Answer: Early in your career, explore broadly to find what you enjoy. After one or two years, specialise enough to be clearly good at something, because employers hire for specific problems. Later, add breadth again as you move into senior or lead roles.

Growth and Career Change

Question: How do I switch from a non-technical job to technology?

Answer: Pick one target role, learn its core skill stack in a structured order, build three portfolio projects, get one recognised certification if the field values it, and use the domain knowledge from your current industry as an advantage. For example, an accountant moving into data analysis already understands finance data better than most beginners.

Question: How do I move from IT Support to Network Engineer?

Answer: To move from IT Support to Network Engineer, learn networking fundamentals and subnetting, study for the CCNA, take on any networking tickets available in your current job, build a home lab or use a simulator, then apply for NOC or junior network roles internally first, since internal moves are usually easier.

Question: How do I move from software development to AI engineering?

Answer: To move from software development to AI engineering, keep your existing engineering strengths, then learn machine learning basics, then learn how to use language models through APIs, embeddings and vector databases, then build and deploy two or three AI features. Your deployment and code quality skills are already a strong advantage.

Question: What does career progression look like in technology?

Answer: A typical path is junior or associate for the first two years, then mid-level engineer for roughly two to five years, then senior engineer. After senior, you can choose a technical track towards staff engineer, principal engineer or architect, or a management track towards team lead, engineering manager and director.

Question: How can I keep my skills current?

Answer: Follow release notes and documentation for the tools you use, read a small number of high-quality newsletters or papers regularly, rebuild a small project each year with newer tools, take one structured course a year, and teach or write about what you learn, since explaining something is the fastest way to find gaps in your understanding.

Question: Is AI going to replace technology jobs?

Answer: AI is changing tasks more than it is removing whole roles. Routine coding, first-line support and basic reporting are being automated, while demand is growing for people who can design, integrate, secure and supervise AI systems. The practical response is to learn to use AI tools well and to deepen skills in judgement, system design and communication, which are harder to automate.